

Confirmation Bias and the Catch-22 of State-building: Evidence from the 2014-2015 Ebola Crisis in Liberia

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Abstract

This paper proposes that confirmation bias — the tendency for individuals to let their prior beliefs and expectations inform how they process new information — can undermine efforts to build political trust and engender citizen compliance in fragile states. In these settings, citizens who mistrust their government and exhibit confirmation bias will be reluctant to credit government for good performance and quick to punish it for bad performance, making it especially difficult for governments to build trust through service provision; they will also be resistant to efforts to engender voluntary compliance through persuasion, making it especially difficult for governments to uphold the law. Together, these tendencies make it harder for governments to govern and harder for them to legitimate their authority with what they manage to achieve. I provide support for these hypotheses using panel and survey-experimental data collected during the 2014-2015 Ebola crisis in Liberia.

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1 Introduction

In weak states, mistrust of government is both a common phenomenon and a key impediment to successful state-building. Because citizens who mistrust their government are unlikely to comply with its demands (Levi, Sacks and Tyler 2009), a lack of trust can make it all but impossible for authorities to raise revenue, uphold the rule of law, or provide public goods. At its most extreme, mistrust can lead to crises of legitimacy and social unrest, threatening the stability of political regimes (Bates 2008; Nunn, Qian and Wen 2018; Rotberg 2004).

Mistrust of government has proven to be remarkably persistent overtime, even in countries experiencing unprecedented improvements in living standards and governance (Letki 2018). Indeed, research on the long-term effects of historic events suggests that negative shocks to trust can endure for decades or even centuries, with important downstream effects on health, wealth, and political behavior (Becker et al. 2016; Lowes and Montero 2018; Nunn and Wantchekon 2009).

In this paper, I propose that one factor that contributes to the persistence of mistrust in weak states is the tendency for individuals to let their prior beliefs and worldviews influence how they interpret politically-relevant information. Underlying this hypothesis is the idea that political affairs are “inevitably complex and therefore often subject to alternative interpretations” (Berinsky and Kinder 2006, p. 641). Faced with complex and ambiguous information about government, citizens with limited time, bandwidth and cognitive capacity tend to rely on heuristics, and these heuristics systematically preference prior beliefs, expectations, and worldviews (Nickerson 1998; Tversky and Kahneman 1974).

I focus on two particular channels through which so-called confirmation bias may impede state-building. The first concerns how citizens evaluate evidence of government performance and use these evaluations to update their beliefs about government trustworthiness. Because these evaluations are complex and often subjective, citizens are likely to interpret signals of performance in whichever manner is most consistent with their prior beliefs. As a result, initially mistrustful citizens will be less likely to credit government for public goods and services and more likely to blame government for negative events, as compared to those who are relatively trusting. (Bayesian models, by contrast, predict the opposite — mistrustful individuals should update more in response to ‘new’ evidence of good government performance and less in response to ‘old news’ about bad government performance that only confirms their priors). The second channel concerns how citizens evaluate the veracity of rumors and conspiracy theories relative to information from government. Here, I propose that confirmation bias makes mistrustful citizens less receptive to government persuasion and more susceptible to rumors and conspiracy theories that lead to misunderstandings about the merits of voluntary compliance. Together, these tendencies undermine voluntary compliance, making it harder for governments to govern, and harder to legitimate their authority with what they manage to achieve.

I provide support for this argument using panel survey data collected during and after the 2014-2015 Ebola crisis in Monrovia, Liberia. The epidemic provides a useful lens through which to analyze the dynamics of government trust and citizen compliance because it brought citizen-state relations into stark relief, heightening both the demands authorities made of citizens and the services they were obliged to provide to combat the epidemic. To prevent the disease from spreading, the Government of Liberia enacted a vari-

ety of onerous control policies, including a curfew, a ban on public gatherings, and a requirement that all persons displaying signs of Ebola (which presents much like many other common illnesses) be reported to authorities. These policies were initially met with widespread resistance, as mistrust rooted in decades of corruption and abuse led many to believe Ebola was a hoax or a ploy by the government to generate more aid funding (ICG 2015). Compliance improved overtime, but not before the easily-preventable virus was able to spiral out of control.

My first set of findings concerns the link between confirmation bias, belief in misinformation, and compliance during the epidemic. I first show that those who voted for Liberia’s main opposition party in 2011 were significantly less trusting of government prior to the Ebola crisis. Using 2011 vote choice as proxy for pre-crisis mistrust, I proceed to show that during the crisis, opposition supporters were much more likely to believe Ebola was a hoax, to report resisting control policies, to know Ebola victims, and conditional on knowing an Ebola victim in month t , to know a victim in month $t + 1$. Because Ebola spread primarily through contact with victims outside of treatment centers (Chowell and Nishiura 2014), in peoples’ homes and neighborhoods, I interpret these patterns as evidence of non-compliance with social distancing policies by citizens who believed in rumors and conspiracy theories and consequently misunderstood the nature of the disease. Consistent with this interpretation, these patterns obtain only during the early stages of the crisis, when it was still unclear whether Ebola was a real disease, and fade entirely by the later stages of the crisis.

Because these results concern differences overtime, it is unlikely that they reflect inherent differences in health, health-seeking behavior. As further evidence against this alternative explanation, I show that outside the crisis, opposition supporters are no more or less likely to become sick from common illnesses or to access health care when they become sick. It is also unlikely that these patterns reflect differences in access to information, as Monrovia’s media environment was flooded with Ebola messaging throughout the crisis.

My second set of findings concerns the relationship between prior beliefs, confirmation bias, and how citizens update in response to evidence of government performance during the later stages of the epidemic. Exploiting plausibly exogenous variation in exposure to government and NGO outreach during the crisis (evidence of ‘good’ government performance), and plausibly exogenous variation in exposure to hardships during the crisis (evidence of ‘bad’ government performance), I show that relative to the rest of the population, opposition supporters update less positively in response to good performance and more negatively in response to bad performance — the exact opposite of what would be predicted by Bayesian models that assume information is unambiguous.

As a final piece of evidence, I replicate these results through a survey experiment in which respondents were randomly primed to consider either hardships experienced during the epidemic, recent experiences with government service provision, or a neutral topic prior to reporting their perceptions of government trustworthiness. Consistent with the idea that prior trust conditions how individuals process information, opposition supporters were significantly more likely to update negatively in response to both the hardship and services primes.

A large literature in political psychology interprets partisan differences in information processing as evidence of “motivated reasoning” by citizens willing to misinterpret evidence in order to affirm their par-

tisan identity, even if this means holding accurate beliefs (Kunda 1999; Taber and Lodge 2006). While this model of information processing has yielded many important insights into political behavior, I view it as an unlikely explanation for the results presented in this study because during the Ebola crisis, individuals had strong incentives to hold accurate beliefs that likely outweighed motivational goals, and because political parties are new and weakly institutionalized in Liberia. Instead, I view my findings as more consistent with research in psychology suggesting confirmation bias occurs when rational but constrained individuals adopt heuristics in the face of complex and uncertain information (Nickerson 1998).

Taken as a whole, my results support the hypothesis that confirmation bias influences how citizens interpret new information, and that this tendency can make it difficult for governments to build trust through service provision or engender voluntary compliance through persuasion. While intuitive, this finding has important implications for the literature on state-building, which has largely overlooked the possibility that citizens are biased learners. Central to this literature is the idea that governments build trust and legitimate their rule by providing public goods and services (Lake 2010; Lipset 1959). Many have argued that even in states with low baseline levels of capacity, modest improvements in performance can lead to greater trust, greater compliance and further improvements in performance, precipitating a virtuous cycle that propels states along the path to development (Englebert 2002; Gilley 2009). My findings suggest this model of political learning may be somewhat optimistic: if mistrustful citizens systematically under-react to evidence of government performance and overreact evidence of government failure, then mistrust is likely to persist overtime, even as government performance improves.

This study also contributes to the literature on citizen cooperation during public health epidemics. Prior research has shown that trust in government and trust in medical providers in particular can shape individuals' decisions to comply with prevention policies (Grossman, Philips and Rosenzweig 2015; Jegede 2007; Martinez-Bravo and Stegmann 2017), and that historical events that undermine such trust can have enduring legacies on health and health-seeking behavior that manifest during epidemics (Lowes and Montero 2018; Thomas and Quinn 1991). These studies largely view trust as influencing behavior 'directly', in the sense that individuals resist cooperation because they mistrust the intentions of healthcare providers. In this paper, I highlight how mistrust can also influence behavior indirectly, through its effect on how individuals evaluate rumors, conspiracy theories, and other forms of misinformation that lead to misunderstandings about the nature of threats to health.

The following sections situate this study in the broader literature on state-building, outline my argument in greater detail, and introduce the data and research design. After presenting the results, I conclude with a discussion of the implications of my findings.

2 Political trust and government performance

The belief that government is trustworthy reflects the judgment that authorities are motivated to act in the public interest and are capable of doing so (Levi, Sacks and Tyler 2009, p. 356). A government's reputation for trustworthiness in the eyes of its citizens is an important determinant of its performance capacity because a strong reputation promotes legitimacy and voluntary compliance (Levi, Sacks and Tyler 2009),

thereby helping governments rule efficiently and “without constant recourse to coercion and without an outlay of resources that would cripple [them] or prevent [them] from providing public goods” (Migdal 2001, p.131). Whereas a trusted government is likely to enjoy widespread voluntary compliance, allowing it to focus scarce resources on public goods provision, a government that lacks the public’s trust will have to spend disproportionate amount of its scarce resources on monitoring and enforcing compliance, leaving little leftover for public goods provision.

Empirically, researchers have documented the importance of political trust for compliance and government performance across a wide range of domains and settings. In the U.S., studies show that those who trust their government are more likely to cooperate with the police (Tyler 2006), pay their taxes (Scholz and Lubell 1998), serve in the military (Levi 1997), comply with vaccination campaigns (Marlow, Waller and Wardle 2007), and support policies in the public’s long-term interest over those that provide short-term individual benefits (Hetherington 2005; Theiss-Morse, Barton and Wagner 2015). Though data on trust and compliance outcomes in developing countries is more limited, existing studies largely confirm the essential role that trust plays in promoting compliance and government efficacy (Bodea and LeBas 2014; Gilley 2006; Keele 2007).

Despite its importance for government performance, political trust has proven remarkably elusive in many of today’s weakest states (Hutchison and Johnson 2011; ?). Scholars have identified a number of factors that may contribute to this persistence, including corruption (Rose-Ackerman 2001), the absence of democratic institutions (Przeworski, Stokes and Manin 1999), and historical legacies (Acemoglu et al. 2014; Nunn and Wantchekon 2009), but arguably the most fundamental explanation is what Englebert (2002) refers to as “the catch-22 of creating legitimacy out of efficiency” in states with low initial levels of capacity (p. 100). The problem, as Englebert explains, is that “legitimacy [trust] in time $t + 1$ results from the efficiency displayed in time t . But successful implementation of development policies in time t requires a certain preexisting level of legitimacy [trust]” (Englebert 2002, p.100). The result is a chicken-and-egg problem that can leave states trapped in situation of perpetual weakness, unable to build capacity because they lack a trustworthy reputation, and unable to improve their reputation because they lack the capacity to perform.

In this paper, I propose a complementary explanation for the persistence of mistrust rooted in psychological theories of information processing. My argument is founded on the idea that political affairs are inherently complex and often ambiguous, and therefore often subject to multiple interpretations. Drawing on theories of information processing, I suggest that when individuals’ encounter political information that is complex and ambiguous, they exhibit *confirmation bias* — they use their prior beliefs, expectations, and worldviews to inform how they should interpret this new, ambiguous information. This tendency, in turn, can undermine efforts to build trust and engender compliance in fragile, illegitimate states through two distinct channels. First, confirmation bias makes initially mistrustful citizens resistant to efforts by government to engender compliance through persuasion and susceptible to rumors and misinformation that undermine the merits of compliance. Second, because attribution of credit for government services is often ambiguous, confirmation bias makes mistrustful citizens less likely to credit government for good performance and more likely to punish it for bad performance. This, in turn, makes it harder for governments to build trust with what they manage to achieve. The remainder of this section outlines my argument in further detail.

Confirmation bias in the face of uncertainty

Political affairs unfold in a world characterized by inherent uncertainty, fundamental complexity, and imperfect information (North 2008, p. 24). Even in the developed democracies, where the majority of voters are well-educated and able to access credible media, national and world affairs are often too big and too complex for citizens to easily comprehend (Berinsky & Kinder, 2006).

A central tenet of information processing theory holds that, when faced with information that's of uncertain meaning or credibility, individuals have a tendency to exhibit confirmation bias — they interpret such information in a manner that's consistent with their prior beliefs, expectations, and worldviews (Nickerson 1998). Though there are a wide range of psychological processes that can lead to confirmation bias — far more than can be summarized here — psychologists generally view them as falling into either of two broad categories (Evans 1989; Kunda 1990; Tversky and Kahneman 1974). The first and perhaps most common category encompasses heuristics adopted by 'unbiased' individuals who value 'truth' but lack either the ability or time required to 'accurately' interpret information (Baron 2014; Bullock 2009; Nickerson 1998). One example of heuristic reasoning that can lead to confirmation bias is the use of partisan cues to assess source credibility or interpret the meaning of political events (Bullock 2011; Nyhan and Reifler 2010). When messenger partisanship is used to infer the credibility of the message, individuals with different prior beliefs who receive the same information may interpret it in very different ways, leading to polarization. Fryer, Harms and Jackson (2015) propose an alternative model of confirmation bias that occurs when ostensibly unbiased agents use Bayes' rule in an iterative way when presented with ambiguous signals about the world — first to interpret the signal and then to form a posterior belief. When agents with different priors interpret signals in a different way before updating, the result is confirmation bias and polarization.

The second driver of confirmation bias highlighted in the literature on information processing is so-called "motivated reasoning" by agents who value their prior beliefs, identities, and world-views. To preserve these attachments and avoid cognitive dissonance, they will misinterpret or ignore evidence against their preferred beliefs and more readily accept evidence that supports them. As with heuristic reasoning, the potential for motivated reasoning increases with the ambiguity of the signal, for while "agents are motivated to arrive at particular conclusions, they are also motivated to be rational" (Kunda 1999, p. 102). In effect, ambiguity provides the motivated reasoner with an out, allowing one to adopt a preferred interpretation while maintaining the guise of rationality.

When thinking about the implications of biased updating, it is helpful to contrast it with a model of unbiased, Bayesian updating. Here, I follow Bullock (2009) and Gerber and Green (1999) in modeling beliefs about a particular subject — here, a politician's level of honesty — as a normally distributed probability distribution, $N \sim (\hat{\mu}_0, \sigma_0^2)$, where $\hat{\mu}_0$ represents an individual's best guess about the politician's true level of honesty, μ , at time $t = 0$ and σ_0^2 represents their level of confidence in that assessment. New information comes in the form a signal, x , which is assumed to be drawn from a normal distribution centered around the politicians 'true' level of honesty, $x \sim (\mu, \sigma_x^2)$, with variance σ_x^2 capturing how definitive the new information is. Upon receiving this information, unbiased individuals use Bayes' Rule to form a posterior belief, $\hat{\mu}_1 | x \sim N(\hat{\mu}_1, \sigma_1^2)$, where, following a common result in statistics,

$$\hat{\mu}_1 = \mu_0 + (x - \mu_0) \frac{\sigma_0^2}{\sigma_0^2 + \sigma_x^2}$$

$$\sigma_1^2 = \frac{\sigma_0^2 \sigma_x^2}{\sigma_0^2 + \sigma_x^2}$$

In this framework, how much an individual updates her belief in response to new information, $|\hat{\mu}_1 - \hat{\mu}_0|$, is determined by i) the difference between signal, x_1 and her prior $\hat{\mu}_0$, and ii) the strength of her prior, σ_0^2 , relative to the credibility of the signal σ_x^2 . Intuitively, an individual who receives ‘new’ news that is far from her prior will update more than if that news were consistent with what she already believed; similarly, she’ll update more if that news comes from a trusted source than if it comes from a source known for circulating rumors.

Where models of learning under confirmation bias depart from models of Bayesian learning is in relaxing the assumption that new information is always unambiguous and interpreted in the same way by all individuals. Instead, both the meaning of the signal, x_1 , and the credibility of the signal σ_x^2 , is sometimes ambiguous, whether on account of its complexity, inherent ambiguity, or due to incomplete information about its source, opening the door for citizens to formulate their own subjective judgments about the meaning and credibility of news and events. The result is that people who observe the *same* signal can update their beliefs in very different ways, depending on how they (subjectively) interpret the signal, x_1 , and its level of credibility, σ_x^2 .

Confirmation bias, trust, and compliance

In this paper, I focus on confirmation bias at two key stages of the “feedback loop” between trust, compliance, and government performance (Gilley 2009). The first is the link between trust and compliance. Here, citizens face complexity when trying to learn about the benefits of compliance as they decide whether to voluntarily comply with government demands. The decision to comply, as Levi (1997, p. 27) argues, “does not occur in a social and political vacuum,” but is instead informed by “personal observations and information provided by acquaintances, media, and organizations.” This discourse will provide a diverse array of perspectives about the merits of compliance, much of which will be ambiguous and ‘open to interpretation’ due to its complexity, doubts about its credibility, or uncertainty about the role of government actors. In the face of this complexity, individuals who are initially mistrustful of government are likely to interpret new information in a manner that undercuts the merits of voluntary compliance, thus making it more difficult for government to govern.

As an example of how this process might play out in practice, consider the U.S. government’s effort to compel citizens sign up for health insurance under the Affordable Care Act (ACA). The roll-out of this reform was plagued by rumors and misinformation, including the allegation that the act provided for the creation of “death panels” with the authority to “pull the plug on Grandma” (Berinsky 2015). Despite efforts by the U.S. government to counter misinformation and build support for the bill through outreach and awareness, rumors about the ACA were widely accepted among Republicans (Berinsky 2015), who were

also much more likely than Democrats to abstain from signing up for medical insurance (Lerman 2018). One potential explanation for these patterns would be that Republicans' distrust of government made them more likely to accept rumors surrounding the ACA, which in turn undercut their assessment of the benefits of compliance with the ACA and contributed to their lower rates of compliance.¹

Examples of how this dynamic may play out in other settings are many. Efforts by governments to coordinate behavior at scale — conscription into the military, taxation, health campaigns, census's, etc. — are almost always accompanied by government outreach and propaganda, as well as by vocal skepticism, rumors, and at times outright misinformation in the media and social discourse. In these situations, prior mistrust has the potential to undermine compliance both directly, through its effect on perceptions of government competence and legitimacy, as well as indirectly, through its effect on information processing and subsequent (mis)understandings of the merits of compliance.

Confirmation bias, trust, and government performance

Learning about the benefits of compliance with government is not the only time that citizens encounter complex and often ambiguous political information — they also face complexity when updating their beliefs in response to evidence of past government performance (Healy and Malhotra 2013). This is because the signals that citizens receive about government performance — the construction of road or school, for example, or the provision of care at a local health clinic — are almost always subject to some degree of ambiguity on account of uncertainty about how to attribute such performance and what should be expected from government to begin with. Particularly in developing countries, where non-governmental organizations play a prominent role in service provision and citizens have limited information about the functioning of government, evaluations of government performance and thus legitimacy are likely to depend on factors such as prior expectations of what the state should and can provide; whether success is attributable to the state relative to non-state actors; and whether the delivery of a particular service is indeed part of the social contract, among other subjective and often unknowable factors (Harding 2015; McLoughlin 2014).

Theories of information processing suggest that when citizens encounter ambiguous signals about the government's past performance, they tend to rely on their prior beliefs and expectations to inform their interpretation. Thus, when a mistrustful individual encounters evidence of good government performance, she is liable to assume that credit goes to luck or NGO's, or that services should have been of even better quality given the government's resources. When she encounters evidence of government failure, she is liable to assume the government is blameworthy, rather than NGOs or other forces beyond government control. The result is biased updating, with mistrustful citizens updating less in response to evidence of good performance and more in response to negative performance, relative to more trusting citizens. Notably, this prediction is the exact opposite of what would be predicted by Bayesian models of updating that assume signals are unambiguous — because good government performance is 'new' news and government failure is 'old' news, initially mistrustful citizens should update more in response to good news and less in response

¹Note that the hypothesis here is that mistrust negatively affects compliance indirectly, through its effect on information processing. This of course does not preclude that mistrust also had a direct, negative effect on compliance (e.g. perhaps Republicans were less likely to believe that public insurance pools would be well-managed).

to bad news.

3 Setting

This study provides support for these predictions using data collected during the 2014-2015 Ebola epidemic in Liberia. Though the epidemic was labeled a health crisis, it was also a *governance crisis* characterized by widespread non-compliance with controversial yet essential control policies (ICG 2015).

From a research perspective, the epidemic provides a unique opportunity to study the dynamics of the social contract because it brought citizen-government relations into stark relief, heightening both the demands authorities made of citizens and the services they were obliged to provide to combat the epidemic.² Under the state of emergency, citizens were required to comply with quarantines, case reporting and contact tracing; they were instructed not to care for loved ones suffering from Ebola-like symptoms, and not to wash the bodies of the dead prior to burial (as is customary). At the height of the epidemic, the government banned public gatherings and imposed a 6 PM curfew, and handshaking, kissing, and touching were strongly discouraged. Collectively, these measures had a profound impact on citizens' everyday patterns of activity and marked an unprecedented attempt by government to exert authority over the lives of citizens.

Initially, these policies were met with skepticism (Abramowitz et al. 2015). Ebola was mysterious and not yet widespread, and preventative measures were costly, inconvenient, and contravened basic human instincts to care for the ill and honor the dead. Compounding matters, mistrust of government rooted in years of corruption and abuse of power led many to believe that Ebola was a scam to attract donor funding. By most accounts, widespread non-compliance with control measures and non-reporting of cases was the main driver of the rapid rise of Ebola in August and September, 2014 (Chowell and Nishiura 2014).

As the epidemic intensified, the government of Liberia and its international partners initiated an ambitious public awareness campaign to combat rumors and persuade citizens to comply control policies. The campaign deployed teams of volunteer health workers into communities to educate citizens about the signs and symptoms of Ebola, demonstrate proper preventative techniques, disseminate information on where to get care, and encourage compliance with control policies. In the later stages of the epidemic, when the majority of resident understood and accepted the reality of the disease, the campaign focused on maintaining vigilance and sustaining practices such as hand-washing and avoiding unnecessary contact.

While the campaign is widely credited with helping to turn the tide of the epidemic, it did not come in time to prevent the epidemic from wreaking havoc on Liberia's economy and health care system (Bowles et al. 2015; Chan 2014). At the peak of the epidemic, most employers, government agencies and local NGOs ceased operations, leading to food shortages and a dramatic rise in unemployment. Many hospitals and health clinics were forced to close as health workers stayed home for fear of contracting the virus, forcing many citizens to forego treatment for illnesses unrelated to Ebola (Morse et al. 2016).

Liberia was finally declared Ebola free in June 2015, more than fifteen months after it registered its first case. Over the course of the epidemic, Liberia registered more than 10,666 confirmed or suspected cases,

²In leveraging the unique circumstances of a crisis event to study citizen-state relations, my study follows in the footsteps of Levi (1997), Lieberman (2009), and Gourevitch (1986).

4806 of which resulted in death. Though the epidemic lasted for over a year, the vast majority of cases (over 80 percent) were registered in just three months — August, September, and October, 2014 (WHO 2015).

Predictions

My argument provides two sets of predictions regarding citizens' beliefs and compliance behaviors during the Ebola epidemic. The first relates to how prior beliefs about government trustworthiness influence information processing as citizens learn about the value of compliance and decide whether to voluntarily comply. During the outbreak, citizens were inundated with information about Ebola. In addition to official sensitization messages from the government and NGOs, citizens encountered information about Ebola on radio talk shows and in social settings. This discourse provided a diverse array of perspectives about the value of compliance with regulations enacted to combat the epidemic – some claimed Ebola was a hoax, neither real nor lethal, others suggested it was a real disease and lethal, but that its spread was being facilitated by health workers, and still others endorsed the position of the government and NGOs.

I expect that citizens who were mistrustful of government at the outset of the outbreak will exhibit confirmation bias when processing this information, and thus to be more likely to have believed rumors suggesting Ebola was a hoax or a government conspiracy, as compared to citizens who were more trusting at the outset of the epidemic. As a result of these misunderstandings, they should be less likely to voluntarily comply with preventative measures and social distancing policies, more likely to become infected, and more likely to spread the disease to others, effectively limiting the government's capacity to control the epidemic.

The second set of predictions relates to how citizens update their beliefs in response to evidence of good and bad government performance during the epidemic. Here, I expect citizens who were mistrustful of government at the outset of the outbreak to be less likely credit the government for services provided during the epidemic, such as community outreach, and thus less likely to positively update their beliefs about government trustworthiness. By the same logic, I expect mistrustful citizens to be quick to blame the government for hardships resulting from the epidemic (such as witnessing dead bodies, knowing Ebola victims, or losing a job) and thus more likely to negatively update their beliefs about government trustworthiness.

4 Research design

My research design employs panel data collected from a representative sample of over 1500 adult residents of Monrovia, Liberia. The first wave of the survey was conducted in December 2014, roughly two months after the height of the epidemic but still a period of active transmission. The second wave was administered to a random subset of respondents from the December survey in March 2015, when there were only a handful of active cases in Liberia; the third and final wave was administered to the same random subset of respondents in June, 2015, shortly before Liberia was officially declared Ebola free by the WHO.

The surveys collected information about belief in misinformation and compliance with health regulations in the initial months of the epidemic, suspected cases within respondents' social networks between July and December, exposure to hardships during the epidemic, trust in government at the time of the survey, and socio-economic and demographic background information.

Because the panel does not cover the pre-crisis period, I cannot directly compare respondents who were mistrustful of government before the crisis to those who were relatively trusting. The survey did, however, inquire about vote choice in the 2011 Presidential election, allowing me to compare those who voted for the main opposition party in 2011 to those who either voted for the incumbent or did not vote at all. In Liberia, the main opposition party, the Congress for Democratic Change (CDC), is a populist party that draws support from those who feel marginalized by the government and mistrust political elites. Known in particular for their opposition to “the resource-grabbing and wealth keeping of government” (Bøås and Utas 2014, p. 58), the CDC and its supporters have a long history of antagonism with the state, dating back to the party’s inception in Liberia’s 2005 election, which the CDC maintains it lost because of fraud (ICG 2012). Mistrust among supporters of the CDC became particularly severe in the aftermath of the 2011 Presidential election, when allegations of fraud led to clashes with security forces and an opposition boycott of the run-off. Though the election was ultimately declared free and fair by domestic and international observers, supporters of the CDC continued to feel as though they had been “cheated by a political elite far away from them and their fragile realities,” a perception which only served to “add to their negative feelings about the state and the democratic project” (Bøås and Utas 2014, p. 60).

Afrobarometer data collected in July 2012, eighteen months before the onset of the epidemic, provides support for the idea that supporters of the CDC were much less trusting of government than the rest of the population prior to the epidemic. The survey asked respondents about their perceptions of seven government institutions — the Presidency, the legislature, the army, the revenue authority, the electoral commission, the police, and the courts. On average, opposition supporters are 14 percentage points less likely to say they trust these institutions and 19 percentage points more likely to say they are corrupt, as compared to the rest of the population (Table 1).

Apart from their mistrust of government, supporters of the opposition in 2012 were broadly similar to the rest of the population on many factors that might influence their beliefs and behaviors during the crisis, including wealth, education, employment status, and trust in neighbors and relatives (Appendix A.3). Though there are some differences — supporters of the opposition are younger, less likely to be female, and somewhat less likely to be Muslim than the rest of the population — these differences are small relative to the gap in political trust.

These considerations suggest that the predictions developed above regarding how mistrustful individuals might differ from the rest of the population in their beliefs and behaviors during the epidemic should apply to supporters of the opposition as well — if they’re prone to confirmation bias, they should be more likely to have believed rumors about Ebola, less likely to have complied with health regulations, and more likely to contribute to the spread of Ebola than the rest of the population. They should also be reluctant to positively update their beliefs in response to evidence of good government performance, and quick to downgrade their beliefs in response to evidence of bad performance.

Table 1: Support for the opposition and trust in government

	Opposition Supporter	Does not support opposition	Controlled Difference
Political trust			
Trust President	0.60	0.36	-0.23**
Trust legislature	0.44	0.31	-0.11*
Trust electoral commission	0.42	0.24	-0.16**
Trust revenue authority	0.33	0.23	-0.10*
Trust police	0.29	0.10	-0.16**
Trust courts	0.43	0.38	-0.03
President is corrupt	0.33	0.59	0.23**
Parliament is corrupt	0.46	0.62	0.12
Govt officials are corrupt	0.46	0.69	0.17**
Police are corrupt	0.55	0.83	0.27**
Revenue authorities are corrupt	0.55	0.73	0.17**
Courts are corrupt	0.44	0.59	0.14**

Notes: Results based on Round 5 Afrobarometer Survey data collected in 2012. Controls include: gender, age, employment, education, religion, and wealth. Robust standard errors in brackets. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Data

The December survey was conducted between December 6th, 2014, and January 10th, 2015 via face-to-face interviews with Liberian enumerators. At the time, Liberia was still under a state of emergency and many of the regulations enacted to curb the epidemic were still in place, but the virus was largely contained, with only 400 suspected or confirmed cases reported in all of December (by comparison, 2743 cases were reported in September, the peak of the epidemic). Details on the sampling procedure and precautions taken to ensure the safety of enumerators are discussed in the Appendix.

In March 2015, 765 respondents from the December survey were randomly selected for a follow-up phone survey, 605 of which were successfully re-surveyed. In June, 800 respondents from the December survey were randomly selected for follow-up, 583 of which were successfully re-surveyed.³ Because they were designed to track trends in trust, health, food security, and economic well-being, the follow-up surveys were much shorter than the December survey, and did not collect data on compliance behaviors, knowing Ebola victims, or hardships. For this reason, the analyses below use only the December survey, unless otherwise noted.

³Analyses exploiting the panel structure of the data weight by the inverse probability of attrition to account for possible non-random attrition.

Outcome variables

To measure belief in rumors and misinformation, respondents were asked to reflect on their beliefs “during the time of Independence Day on July 26” — about a month before the height of the outbreak — and to report i) whether they believed Ebola was a “real disease” that could kill them, and ii) whether they thought Ebola was “a scam by the government to attract donor funding.” To measure compliance at the height of the epidemic, respondents were asked to retrospectively report whether they were willing report cases or provide health workers with information about dead bodies around the time of Independence Day. Though these questions are subject to response bias, they were not sensitive during pretesting (and indeed, affirmative responses were quite common in our survey, as shown below).

Table 2: Descriptive statistics

	Mean	N
Beliefs and compliance		
Believed Ebola was a scam?	0.38	1557
Believed Ebola was not a real disease?	0.69	1557
Willing to report cases?	0.66	1557
Willing to allow health workers access to dead bodies?	0.59	1557
Knew a victim of Ebola?	0.29	1557
Trust in government		
Trust government?	0.24	1557
Trust Ministry of Health?	0.27	1557
Trust police?	0.20	1557
Government is corrupt?	0.73	1557
MOH is corrupt?	0.68	1557
Police are corrupt?	0.76	1557
Government is able to provide good education?	0.28	1557
Government is able to provide security?	0.41	1557
Government is able to provide quality healthcare?	0.35	1557
Government has the heart to provide healthcare?	0.15	1557
Government has the heart to provide security?	0.20	1557
Government has the heart to provide education?	0.13	1557
Outreach		
Government outreach	0.39	1557
NGO outreach	0.65	1557
Hardships		
Witnessed dead bodies?	0.22	1557
Lost job in past six months?	0.47	1557
Family member sick and forewent care, past 3 months?	0.20	1557
Family member lost job in past six months?	0.44	1557

For obvious reasons, it is not possible to measure Ebola’s spread overtime by asking respondents whether they were ever infected with Ebola.⁴ The approach taken here is to measure Ebola’s spread indirectly, by asking respondents whether any of their friends, family members or neighbors were infected with Ebola, and if so, in which month. If these individuals are also less trustful of government and less likely to comply with

⁴Indeed, as a safety precaution, enumerators avoided all households that ever had a suspected case.

preventative measures, opposition supporters should be more likely know a victim of Ebola, particularly at the height of the epidemic, when the origin and severity of the virus was still in doubt. They should also be more likely to report knowing a victim in month $t + 1$, conditional on knowing a victim in month t .

The second set of outcomes relates to how citizens update their beliefs about government trustworthiness in response to evidence of good and bad government performance. The main outcome for this analysis is an index of government trustworthiness, constructed from sixteen questions about the perceived capacity and intentions of the Ministry of Health, the police, and the government in general. Responses to each question are converted from a Likert-scale to indicators that take a value of 1 for “Agree” or “Strongly agree” responses; these indicators are then aggregated additively into a standardized index of government trustworthiness.

Empirical strategy

I divide my analysis into two sections. The first considers the relationship between trust, information processing, and compliance with health regulations during Ebola; the second considers the relationship between trust, information processing, and the ability of government to build a reputation for trustworthiness through service provision.

(Mis)trust, confirmation bias, and compliance

To assess whether opposition supporters differed in their beliefs and compliance behaviors during the epidemic, I estimate the following regression equation among respondents of :

$$Y_{ic} = \alpha_c + \gamma \text{OppositionSupporter}_{ic} + X_{ic}^T \beta + \varepsilon_{ij} \quad (1)$$

where Y_{ic} one of the measures of belief in rumors and willingness to comply during the crisis for individual i of community c , and X_{ic} is a vector of covariates.⁵ The inclusion of community fixed effects (α_c) ensures differences are not driven by differences in media exposure or access to health facilities; the inclusion of X_{ic} ensures the results are not driven by differences in education, gender, religion, wealth or other socio-demographic factors.

To assess whether misperceptions about the severity among opposition supporters made Ebola more likely to spread, I adopt a difference-in-differences strategy, comparing contagion within opposition supporters networks to non-opposition supporters during July, August, and September, when the existence of Ebola Virus Disease was widely questioned, to levels of contagion in September, October, and November, when the lethal severity of Ebola was beyond dispute (and individuals took precautions regardless of their beliefs about Ebola’s origins, transmission pathways, or views toward government).

$$\text{KnowVictim}_{ict} = \sum_{t=1}^6 \delta_t M_t + \sum_{t=1}^6 \tau_t (M_t \times \text{OppositionSupporter}_{ic}) + X_{ic}^T \beta + \varepsilon_{ict} \quad (2)$$

⁵Covariates include: education, income, household size, religion, gender, ethnicity, four age cohort dummies, and turnout in 2011.

where $KnowVictim_{ict}$ denotes whether individual i in community c knew at least one victim of Ebola in month $t \in 1 \dots 6$ between July and December, 2014. Interest lies in the overtime comparison of τ_t , the proportion of opposition supporters who knew at least one victim in a given month, to δ_t , the proportion of the rest of the population that knew at least one victim in a given month, overtime. I expect opposition supporters to be more likely to know victims in July, August, and September, but for this gap to close from October onward, when the lethal severity of Ebola was indisputable.

I adopt the same difference-in-differences approach to test whether Ebola was more likely to *spread* within opposition supporters' social networks, comparing the lagged effect of knowing a victim in month t on knowing a victim in month $t + 1$ overtime across opposition supporters and the rest of the population. The idea behind this approach is that, if social networks supportive of the opposition were indeed less likely to comply with social distancing measures, then Ebola should be more likely to spread once someone in the network becomes infected, resulting in a positive association between knowing a victim in month t and knowing a victim in month $t + 1$, as estimated by:

$$\begin{aligned} KnowVictim_{ict+1} = & \sum_{t=1}^5 \delta_t (M_t \times KnowVictim_{ict}) \\ & + \sum_{t=1}^5 \tau_t (M_t \times KnowVictim_{ict} \times OppositionSupporter_{ic}) \\ & + X_{ic}^T \beta + \varepsilon_{ict} \end{aligned} \quad (3)$$

Interest lies in the overtime difference between τ_t , the lagged effect among opposition supporters of knowing a victim in month t on the likelihood of knowing a victim in month $t + 1$, and δ_t , the lagged effect among the rest of the population. I expect there to be a large gap between these estimates in July, August, and September, but for this gap to close overtime, as opposition supporters and their social networks begin to take Ebola more seriously and comply with control policies, such as by avoiding contact with those exhibiting the signs and symptoms of the disease.

The use of the difference-in-differences estimation strategy ensures these results are not driven by latent differences in the likelihood that opposition supporters get sick, as such explanations cannot account for differences overtime. To further ensure these results are not driven by unobserved factors that influence health or health-seeking behavior, Appendix A.3 shows that supporters of the opposition are no more or less likely to become sick, access formal health when sick, have children become sick, or access formal health care for those children, as compared with the rest of the population.

(Mis)trust, confirmation bias, and trust-building

To test for confirmation bias in how citizens update their beliefs about government trustworthiness in response to evidence of good and bad government performance, I exploit variation in exposure to government and NGO outreach during the epidemic (evidence of “good” government performance), as well as variation in exposure to hardships associated with the epidemic (evidence of “bad performance”), such as losing a job, observing dead bodies in the street before removal, or having had a family member forego medical treatment in the past three months because health facilities were closed. Below, I present evidence suggesting

exposure to outreach and hardships was plausibly exogenous to perceptions of government trustworthiness.

As discussed in Section 3, raising awareness and promoting vigilance through door-to-door outreach was among the most important public goods that government provided during the crisis, in addition to coordinating outreach activities by NGOs. However, because NGOs often operated independently of government, and because they often organized and supported government outreach efforts, exposure to these campaigns has ambiguous implications for government trustworthiness: if attributed to government, then it may be seen as a signal of good government performance, but if attributed to NGOs, it may be seen as an uninformative or null signal. Similarly, hardships resulting from the epidemic could be interpreted as either evidence of government failure, or unrelated to government performance, depending on whether blame is attributed to the government, NGOs, uncooperative citizens, or natural forces.

If confirmation bias indeed influences how citizens process these experiences, then my proxy for pre-crisis mistrust, support for the opposition in 2011, should condition the effects of outreach and hardships on trust. To test this prediction, I estimate:

$$Y_{ic} = \delta GovtOutreach_{ic} + \tau(GovtOutreach_{ic} \times OppositionSupporter_{ic}) + X_{ic}^T \beta + \varepsilon_{ict}$$

where Y_{ic} denotes a standardized index of government trustworthiness, and $GovtOutreach_{ic}$ is a binary variable indicating whether respondent i from community c reported that government health workers ever canvassed her household in the past four months. I estimate the association between trust and exposure to NGO outreach and hardships analogously, with the exception that exposure to hardships is measured as an index ranging from 0-3, denoting the number of hardships experienced by the respondent during the epidemic. Descriptive statistics for these variables are presented in Table 2. If prior levels of trust indeed conditions how individuals process signals of government performance, then the positive association between government and NGO outreach should be weaker among opposition supporters ($\tau < 0$). Similarly, the negative association between hardships and trust should be stronger among supporters of the opposition in 2011 ($\tau < 0$).

Exogeneity of outreach and hardships

Here, I present three pieces of evidence to support the assumption that exposure to outreach and hardships was independent of potential perceptions of government trustworthiness. First, in Table 3, I show that these experiences were balanced on many factors that might influence perceptions of government trustworthiness, including religion, participation in the 2011 election, age, and income. Though there is some imbalance, these factors are only weakly predictive of trust, and therefore unlikely to confound the association between outreach and hardships during the epidemic and trust at the time of the survey.⁶ Insofar as unobserved predictors are similar in nature to the observed predictors, they too are unlikely to confound.

⁶See Appendix A.4.

Table 3: Balance

	(1) Hardships Index	(2) Government outreach	(3) NGO outreach
Above median education	0.12 (0.06)*	-0.05 (0.03)	0.01 (0.03)
Above median income (pre-Ebola)	0.10 (0.06)	-0.00 (0.03)	0.07 (0.03)**
Age 18-30	0.18 (0.14)	-0.10 (0.08)	-0.05 (0.07)
Age 31-50	0.27 (0.13)*	-0.09 (0.08)	-0.06 (0.08)
Age 51-65	-0.04 (0.16)	-0.08 (0.09)	-0.03 (0.08)
Above median household size	0.19 (0.07)**	-0.06 (0.03)	-0.02 (0.03)
Muslim	0.02 (0.11)	-0.06 (0.06)	-0.01 (0.05)
Female	0.01 (0.06)	-0.10 (0.03)**	-0.10 (0.03)**
Voted in 2011 election	0.06 (0.08)	0.02 (0.04)	0.04 (0.04)
Voted for opposition in 2011 election	0.14 (0.07)	-0.05 (0.04)	-0.04 (0.04)
Observations	1515	1141	1141
R^2	0.12	0.13	0.17

Notes: Sample sizes vary because questions on government and NGO outreach were added only after 20% of respondents were already surveyed (in the December survey). Robust standard errors in brackets, clustered by community. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

The second test of the unconfoundedness assumption exploits the follow-up survey administered in June, 2015 to 583 respondents randomly selected from the December survey. The follow-up survey included the same module on perceptions of government as the original survey in December. If exposure to hardships and/or outreach during the epidemic was indeed exogenous to perceptions of government trustworthiness, then we would expect to observe a null association between these variables in June (or at least a much weaker association than observed in December, if effects persist but fade overtime).

Table 4: Balance on Trust Index in June

	(1)	(2)	(3)
	Trust in government index		
hardships_index	0.01 (0.05)		
govt_patrol_dum		0.06 (0.13)	
ngo_patrol_dum			-0.00 (0.19)
Observations	398	286	286
R^2	0.33	0.41	0.41

Notes: Sample sizes vary because questions on government and NGO outreach were added only after 20% of respondents were already surveyed (in the December survey). Observations weighted by the inverse probability of attrition. Robust standard errors in brackets, clustered by community. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Indeed, this is essentially what we see in Table 4. Neither hardships nor NGO outreach associate with trust in June, and while there is a small, positive association between government outreach and trust, this association is statistically insignificant (and much weaker than the association observed in December, reported in Section 5, below).

A third and final check considers whether those who expressed high levels of trust in government in the December survey were subsequently more likely to experience outreach in the first three months of 2015, as measured in our March, 2015 survey.⁷ Because outreach during this time period was similar to outreach in earlier periods, this analysis serves as a direct test of our identification assumption, namely that outreach was not targeted towards inherently trusting individuals.⁸ The results, reported in Table ??, reveal only a weak and insignificant association between trust in December 2014 and subsequent exposure to outreach, suggesting that canvassers do not target inherently trusting individuals.

⁷I cannot run this test for hardships because those experiences only occurred during the height of the epidemic, and thus were only included in the December survey.

⁸Indeed, outreach conducted between January and March 2015 appears to have many of the same impacts as outreach conducted between July 2014 and December 2014.

Table 5: Trust in December 2014 does not predict outreach in early 2015

	Government outreach Jan-Mar 2015			NGO outreach Jan-Mar 2015		
Trust.In.Govt.Dec2014	0.03 (0.02)	0.02 (0.02)	0.01 (0.02)	-0.02 (0.02)	-0.03 (0.02)	-0.00 (0.02)
Observations	541	527	527	541	527	527
Controls?	N	Y	Y	N	Y	Y
Community fixed effects?	N	N	Y	N	N	Y
R ²	0.00	0.04	0.29	0.00	0.06	0.34

Notes: *Trust.In.Govt.Dec2014* is a standardized index of perceptions of government trustworthiness, measured in December 2014. Robust standard errors in brackets, clustered by community. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

5 Results

5.1 Confirmation bias, compliance, and the spread of Ebola

Table 6 displays the association between support for the opposition in 2011 and belief in rumors during the epidemic and willingness to comply with controversial health regulations. Opposition supporters were eight percentage points more likely to believe Ebola was a scam by the government to attract donor funds, nine percentage points less likely to believe it was a real, lethal disease, eight percentage points less likely say they were willing to report suspected cases to health workers, and 4 percentage points less likely to say they would allow health workers access to dead bodies. Notably, in results not shown, I find that opposition supporters were not any less willing to say they would be willing to report dead bodies to health workers at the time of the survey, when misinformation about Ebola were no longer credible; nor were they any less knowledgeable about the virus's transmission pathways or symptoms.

Table 6: Support for the opposition in 2011 and belief in rumors during the epidemic, controlled difference

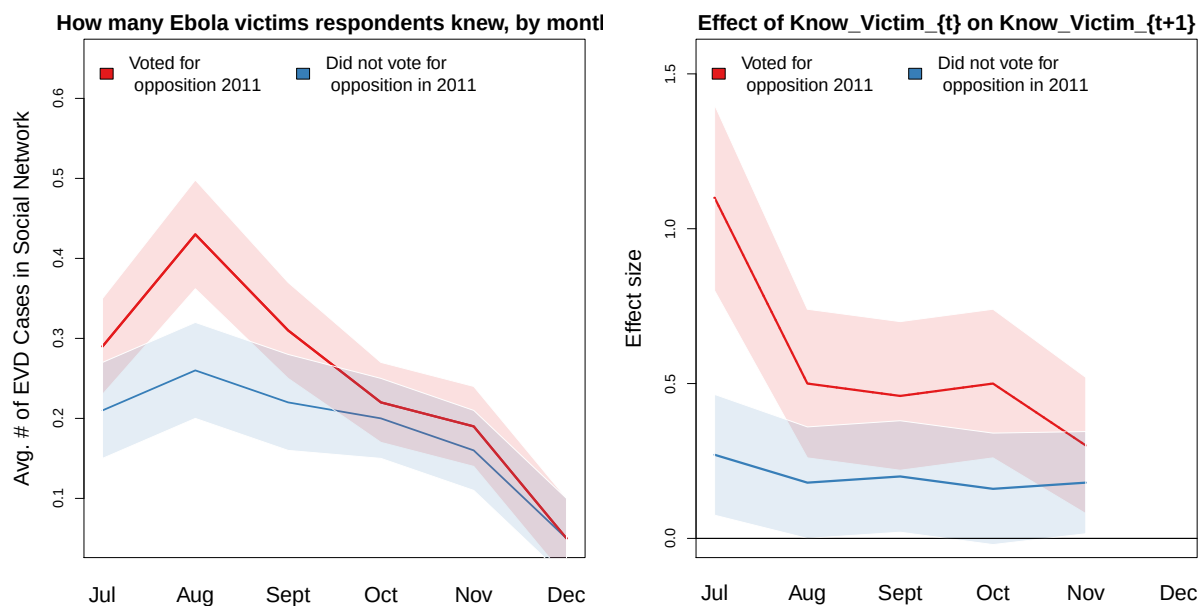
	(1) Believed Ebola was a scam?	(2) Believed Ebola was not a real disease?	(3) Willing to report suspected cases?	(4) Willing to allow health workers access to dead bodies?
Voted for opposition in 2011	0.08 (0.03)**	-0.09 (0.03)**	-0.08 (0.03)*	-0.04 (0.03)
Controls	Y	Y	Y	Y
Community fixed effects	Y	Y	Y	Y
Outcome mean	0.38	0.69	0.66	0.59
Observations	1498	1498	1498	1498
R ²	0.11	0.13	0.13	0.12

Figure 1 provides suggestive evidence that opposition supporters' belief in misinformation during the early stages of the epidemic lead to non-compliance with health regulations and contributed to the epidemic's spread. In particular, supporters of the opposition knew .15 and .12 more victims of Ebola in August and September 2014, respectively, when belief in misinformation was common, as compared with the rest of the population. Yet in the ensuing months, when misinformation was no longer credible, opposition supporters knew about the same number of victims as the rest of the population.

The second panel in Figure 1 provides some insight as to why this might be. It plots the estimated effect of knowing a victim in month t on knowing a victim in month $t + 1$, which I interpret as a crude approximation of the reproductive rate of the disease. Among opposition supporters, knowing a victim in July was associated with knowing 1.44 victims in August, after which point the lagged relationship steadily declines; by contrast, the lagged relationship remains steady at around .3 throughout the crisis among the rest of the population. These patterns suggest that once Ebola entered a social network partial to opposition, it was more likely to spread within that network than if it had entered a social network that was not partial to the opposition.

Because Ebola spread primarily through contact with victims outside of treatment centers (Chowell and Nishiura 2014), in peoples' homes and neighborhoods, it is unlikely that these patterns reflect differences in access to Ebola Treatment Units (ETUs). It is also unlikely that they reflect differences in access to information about Ebola or exposure to rumors, since such information was widespread throughout Monrovia (Abramowitz et al. 2015), and since the analysis includes community fixed effects. Instead, I interpret these results as evidence that supporters of the opposition were less likely to take precautions when a family member or neighbor became infected with the disease.

Figure 1: Mistrust and the spread of Ebola



5.2 Confirmation bias and trust-building

Community outreach was among the most important public services provided by the government and international NGOs during the crisis. In this section, I consider whether exposure to these programs led citizens to positively updated their beliefs about the government's trustworthiness. The results, displayed in Table 7, show that outreach conducted by government canvassers is associated with a .4 standard deviation improvement in perceptions of government trustworthiness across the full sample. The association between NGO outreach and perceptions of government trustworthiness is smaller, at .23 standard deviations, but still statistically and substantively significant. Columns 4 and 6 display how these associations vary according to vote choice in the 2011 election. Though the interaction terms are not significant, both government and NGO outreach appear to be slightly less effective at building trust among opposition supporters, consistent with the hypothesis that opposition supporters are less responsive to evidence of good government performance.

Table 7 also displays the association between hardships experienced during the epidemic and perceptions of government trustworthiness. Substantively, a one standard deviation increase in the hardships index is associated with a .18 standard deviation reduction in trust, consistent with the idea that citizens attributed these hardships to poor government performance, downgrading their beliefs accordingly. However, the negative association between hardships and trust is roughly twice as large among opposition supporters, consistent with the idea that these individuals are more likely to blame government for evidence of poor performance.

Table 7: Heterogeneous effects of outreach & hardships by support for the opposition in 2011

	(1)	(2)	(3)	(4)	(5)	(6)
	Index of government trustworthiness (std)					
Hardships index	-0.18 (0.02)***	-0.16 (0.03)***				
Hardships index x Opposition 2011		-0.08 (0.04)*				
Government outreach			0.40 (0.07)***	0.43 (0.08)***		
Government outreach x Opposition 2011				-0.09 (0.12)		
NGO outreach					0.23 (0.09)**	0.29 (0.10)***
NGO outreach x Opposition 2011						-0.20 (0.13)
Voted for opposition in 2011 election	-0.16 (0.07)**	-0.04 (0.10)	-0.21 (0.08)**	-0.17 (0.09)*	-0.22 (0.08)**	-0.09 (0.12)
Controls	Y	Y	Y	Y	Y	Y
Community fixed effects	Y	Y	Y	Y	Y	Y
Observations	1515	1515	1141	1141	1141	1141
R ²	0.16	0.16	0.16	0.17	0.14	0.14

Notes: Sample sizes vary because questions on government and NGO outreach were added only after 20% of respondents were already surveyed. Robust standard errors in brackets, clustered by community. * $p < 0.1$;

** $p < 0.05$; *** $p < 0.01$.

5.3 Supplemental Evidence

I have presented evidence consistent with the idea that prior perceptions of trustworthiness influence how citizens process information, and that this has consequences for both the ability of government to engender compliance, and its ability to legitimate its rule through service provision. In this section, I present results from two additional empirical tests of this hypothesis.

The first set of supplemental results come from a priming experiment administered to a random sample of 700 adult Monrovia residents in January 2016, roughly six months after Liberia was declared Ebola-free. The priming experiment was embedded in a survey conducted on behalf of an international NGO about residents' attitudes and recovery patterns in the aftermath of Ebola. Immediately prior to soliciting their perceptions of government trustworthiness, respondents were administered either a hardships prime, a government services prime, or a neutral prime (questions about their socio-demographic characteristics). The hardships primed entailed asking respondents about the types of hardships they experienced during the epidemic, such as losing a job, knowing Ebola victims, or witnessing dead bodies. The government services prime entailed asking respondents about whether they had recently received or witnessed services from the government, such as road improvements, infrastructure projects, or vaccination campaigns. A key advantage of this approach to priming is that because the primes are administered discreetly within the flow of the survey, respondents do not actually know they are being experimented on.

Theories of confirmation bias are agnostic about the *average effect* of the government services prime, since that should logically depend on the quality and quantity of actual services that respondents report receiving. Rather, the main prediction is that opposition supporters should be either less likely to positively update their beliefs (if the average effect is positive) or more likely to downgrade their beliefs (if the average effect is negative). Similarly, opposition supporters should be more likely to downward revise their beliefs in response to the hardships prime.

Importantly, opposition supporters in the government services group were not any more or less likely to receive government services, and similarly, opposition supporters in the hardships group were not any more or less likely to experience hardships.⁹ This indicates that opposition supporters were reflecting on the same types of experiences during the prime as the rest of the population, thus ruling out the possibility that priming effects vary because experiences with government services and/or hardships vary.

⁹Opposition supporters reported an average of .84 hardships during the crisis, compared to 0.88 among those who did not vote for the opposition in 2011; for government services, opposition supporters witnessed an average of 1.2 services in the past month, compared to 1.3 among those who did not vote for the opposition in 2011.

Figure 2: Mistrust and the attribution of blame (credit) for negative (positive) performance:

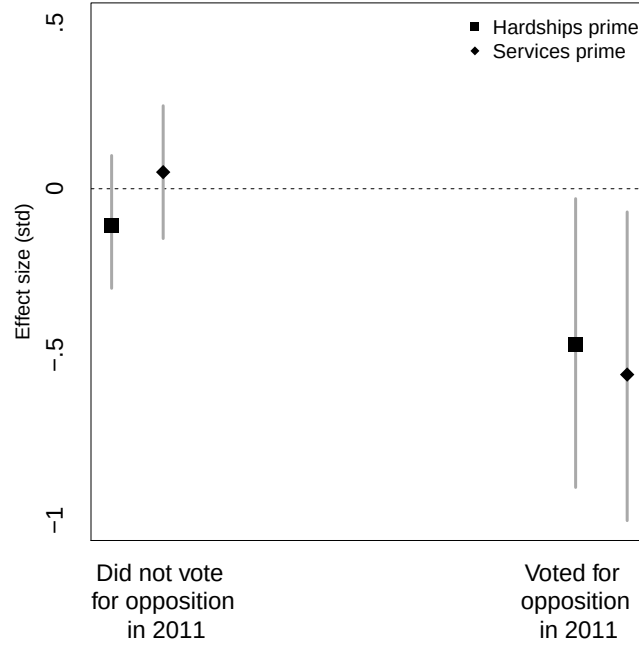


Figure 2 displays the results of the hardships and government services experimental prime. Both primes had a large, negative effect on perceptions of trustworthiness among those who supported the opposition in 2011 but no significant effect among the rest of the population, consistent with the idea that prior beliefs about government trustworthiness condition how citizens process information about government performance in a manner that makes it more difficult for government to build trust among untrusting citizens.

The second piece of supplemental evidence addresses the possibility that support for the opposition in 2011 is an imperfect proxy for pre-crisis trust, and in particular, the concern that differences other than pre-crisis perceptions of government trustworthiness explain why opposition supporters respond differently to evidence of government performance. As a more direct test of the hypothesis that prior *trust* conditions how citizens process evidence of government performance, I exploit the panel structure of the data to test whether trust in December 2014 conditions the effect of outreach in early 2015 on perceptions of trustworthiness, as measured in March of 2015.¹⁰ Formally, I estimate the following difference-in-differences regression:

$$\begin{aligned}
 Trust_In_Government_{it} = & +\delta GovtOutreach_{it} \\
 & +\tau(GovtOutreach_{it} \times OppositionSupporter_i) \\
 & +\alpha Trust_In_Government_{it-1} + X_i^T \beta + \epsilon_{it}
 \end{aligned}$$

where $Trust_In_Government_{it}$ is the standardized index of government trustworthiness for individual i

¹⁰Though NGO and government outreach continued into early 2015, by April of 2015 outreach activities had largely ceased, so use this same strategy to estimate the impact of outreach between March and June 2015 on trust in June.

in March, 2015, $Trust_In_Government_{it-1}$ denotes the same index in December 2014, $GovtOutreach_{it}$ is an indicator for whether respondent i reported that government health workers canvassed her household in the past three months (since the December survey), and all other variables are as defined in Section 4. The results, displayed in Table 8, show that the positive effect of government and NGO outreach is strongest among those who were already trusting of government in December 2014, consistent with the idea that prior trust conditions how citizens process evidence of government trustworthiness. These results should be interpreted with some degree of caution, however, because effect heterogeneity is only significant for NGO outreach.¹¹

Table 8: Trust in December conditions the effect of outreach in early 2015 on Trust in March 2015

	Index of government trustworthiness (March '15)			
Govt outreach (Jan-Mar '15)	0.34 (0.09)***	0.33 (0.09)***		
Trust in Govt (Dec '14) $\times$ Govt outreach (Jan-Mar '15)	0.07 (0.07)	0.09 (0.08)		
NGO outreach (Jan-Mar '15)			0.04 (0.08)	0.04 (0.08)
Trust in Govt (Dec '14) $\times$ NGO outreach (Jan-Mar '15)			0.18 (0.08)**	0.19 (0.08)**
Trust in Govt (Dec '14)	0.17 (0.05)***	0.16 (0.05)***	0.10 (0.06)*	0.09 (0.06)
Observations	541	541	541	541
Controls	N	Y	N	Y
R^2	0.09	0.12	0.07	0.10

Notes: Robust standard errors in brackets, clustered by community. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

6 Discussion

This paper has proposed that citizens' perceptions of government trustworthiness conditions how they interpret new information when deciding whether to comply with government regulations and when updating their beliefs in response to government performance. Drawing on panel data collected during and after Ebola crisis in Monrovia, Liberia, I present three sets of evidence consistent with this hypothesis. First, I show that those who voted for Liberia's main opposition party in 2011 were significantly less trusting of government prior to the Ebola crisis, and that during the crisis, they were much more likely to believe Ebola was a hoax, to report resisting control policies, and to know Ebola victims. Second, exploiting plausibly exogenous variation in exposure to government and NGO outreach during the crisis (evidence of 'good' government performance), and plausibly exogenous variation in exposure to hardships (evidence of 'bad' government performance), I show that opposition supporters update less positively in response to good performance and more negatively in response to bad performance. And finally, I replicate these findings using a survey experiment in which respondents were randomly primed to consider either hardships experienced during the epidemic, recent experiences with government service provision, or a neutral topic, showing that

¹¹One explanation for why effect heterogeneity is stronger for NGO outreach may be that its a more ambiguous signal of government performance, and therefore more subject to confirmation bias.

opposition supporters were significantly more likely to update negatively in response to both the hardship and services primes.

While the idea that citizens exhibit confirmation bias when learning about the state may seem intuitive — and would certainly come as no surprise to scholars of American politics — this hypothesis has far-reaching implications for theories of state-building. In countries where mistrust of government prevails, confirmation bias not only makes it harder for governments to engender compliance (and thus to provide public goods and services), it also makes it harder for them to build trust with the little they manage to provide.

Further research is required to understand whether and under what conditions these findings might generalize. The 2014-2015 Ebola crisis was a unique event, and the patterns reported in this study may not generalize to other contexts. Nevertheless, the key building blocks of the argument — the complexity associated with inferring the ‘true’ meaning of political information, and the reciprocal relationship between political trust and government performance — apply generally.

My results also call attention to the need to better understand the drivers of confirmation bias in weak and illegitimate states. Though scholars of American politics have long studied confirmation bias in developed countries, highlighting in particular the role of affective or emotional attachments to party, ideology, and religion (Healy and Malhotra 2013; ?), the subject has largely been neglected by state-building scholars. One possibility suggested by theories emphasizing ambiguity as a source of confirmation bias is that the prevalence of non-state service provision in weak states contributes to confirmation bias by introducing ambiguity about who should be credited for public services. Indeed, this may be one reason why effect heterogeneity was slightly larger for NGO outreach than for government outreach — whereas government outreach is more clearly attributable to government, and thus less subject to confirmation bias, attribution of credit for NGO outreach is more ambiguous, since citizens could reasonably infer that government helped to coordinate NGOs’ efforts (Cruz and Schneider 2016). Conversely, budget support from donor countries could increase ambiguity or cause unrealistically high expectations about what government is capable of, thereby lowering evaluations of government performance even for services delivered solely by government. These considerations suggest that efforts by donor countries to promote welfare or stimulate state-building through non-state service provision or budget support may be partially offset by a higher incidence of confirmation bias that makes it harder for governments to build trust through service provision (McLoughlin 2014; Sachs 2006).¹²

Another potential source of confirmation bias, familiar to scholars of American politics, derives from partisans’ affective or emotional attachments to their political parties. The past several decades have seen the expansion of competitive democratic elections into many of the world’s weak states, including Liberia. In these countries, opposition partisans may be reluctant to positively update their beliefs about the incumbent government because holding favorable views about the incumbent would be inconsistent with their partisan affiliation, potentially causing cognitive dissonance. This explanation seems unlikely in the current study

¹²Cruz and Schneider (2016) argue the opposite, showing that aid can undermine accountability by allowing incumbents to “credit claim” for non-state services they had no hand in providing. My argument suggests this is likely only in places where trust in government is relatively high, and that aid may have the opposite effect in countries with low levels of trust. In these contexts, donors may face a trade-off between winning hearts and minds or addressing humanitarian needs and building trust necessary for strong states.

because the severity of the crisis created strong incentives for individuals to hold accurate beliefs. But it is entirely plausible that confirmation bias motivated by partisan attachments operates in other developing country settings, limiting the ability of governments to both engender trust through service provision and their ability to persuade opposition partisans to comply with controversial regulations.

Apart from its contribution to state-building theory, this study aims to help make sense of what is undoubtedly one of the most dramatic yet puzzling instances of state failure in recent memory. After more than a decade of peace, stability, and unprecedented improvements to living conditions, why did Liberians remain so mistrustful of their government? What prevented them from learning that their government does not, in fact, wish to spread Ebola? And why did international NGOs feature so prominently in the discourse of distrust? My results suggest that the dominance of non-state service providers and the prevalence of budget support from donors may have hindered trust-building prior to the epidemic by preventing citizens from formulating accurate expectations of their government's capabilities and thus making it difficult for them to credit the government for its accomplishments.

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Confirmation Bias and the Catch-22 of State-building: Evidence from the
2014-2015 Ebola Crisis in Liberia

Online Appendix

A.1 Sampling procedures	2
A.2 Enumerator safety precautions	2
A.3 Balance tests	3
A.3.1 Support for the opposition and socio-demographic characteristics	3
A.3.2 Support for the opposition and health and health-seeking behavior	3
A.4 Sensitivity analysis	4

A.1 Sampling procedures

Within each of the 78 Monrovia neighborhoods sampled for the survey, twenty households were randomly selected within each neighborhood following a random walk procedure. Enumerators began by dividing each neighborhood into blocks with the assistance of a local leader. They then selected four blocks at random. Working with the local leader, they next identified the most central location within each block — typically a town square, water pump or “palava hut” from which paths feeding all parts of the neighborhood originated. Enumerators then randomly selected one path and walked the length of it, marking every 5th household with chalk. Upon reaching the end of one path, they turned left and continued walking until they found another. Finally, they returned to each house, created a roster of all individuals living in the house, and selected one of those individuals at random.

A.2 Enumerator safety precautions

We took extensive precautions to ensure the safety of our survey enumerators. First, enumerators avoided any neighborhood with known active Ebola cases or contacts. Within neighborhoods, enumerators coordinated with local leaders to avoid households with known Ebola victims (past or present), suspected Ebola victims (past or present) or otherwise sick persons (in the present). Enumerators were trained to avoid physical contact and maintain a two-foot distance when interacting with respondents. They also monitored their temperatures daily, and were provided with rubber boots and hand sanitizer as additional precautions.

No adverse events to Parley staff or respondents were reported at any time during the surveys.

A.3 Balance tests

A.3.1 Support for the opposition and socio-demographic characteristics

Table A.1: Support for the opposition and balance on socio-demographic characteristics

	Opposition Supporter	Not an Opposition Supporter	Difference in Means	Tstat
Female	0.60	0.36	-0.23	-4.43
Age	36.67	35.25	-1.41	-1.12
Employed	0.27	0.24	-0.02	-0.49
No education	0.14	0.10	-0.03	-0.94
Primary school education only	0.19	0.17	-0.01	-0.28
Secondary school or above education	0.30	0.33	0.03	0.64
Muslim	0.16	0.09	-0.07	-1.95
Wealth index (std)	0.39	0.31	-0.09	-0.80
Trust relatives	0.56	0.62	0.04	0.73
Trust neighbors	0.34	0.39	0.01	0.17
Trust other people	0.17	0.17	-0.02	-0.54

Notes: Data: 2012 Round 5 Afrobarometer Survey.

A.3.2 Support for the opposition and health and health-seeking behavior

Table A.2: Support for the opposition in 2011 and balance on health and health seeking behavior outside Ebola

	Adult member of HH sick in past month?	Child member of HH sick in past month?	If adult sick: sought care at clinic?	If child sick: sought care at clinic?
Voted for Opposition in 2011	-0.01 (0.02)	0.03 (0.03)	-0.01 (0.05)	-0.03 (0.05)
Observations	2610	1952	531	615
Controls?	Y	Y	Y	Y
R^2	0.01	0.02	0.06	0.05

Notes: Results pooled across all three survey waves (Dec '14, Mar '15, and Jun '15). Robust standard errors in brackets, clustered by community. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

A.4 Sensitivity analysis

This section assesses the robustness of the results presented in Table 7 to omitted variables bias, following the approach proposed in Oster (2017). The idea is to use the bias eliminated by observed covariates to assess the potential bias of unobserved variables. Formally, consider the following linear regression models:

$$Y = \beta X + \gamma \mathbf{W}_1 + \alpha \mathbf{W}_2 + \varepsilon \quad (1)$$

$$Y = \tilde{\beta} X + \tilde{\gamma} \mathbf{W}_1 + \tilde{\varepsilon} \quad (4)$$

$$Y = \hat{\beta} X + \hat{\varepsilon} \quad (5)$$

where β , the effect of some treatment X , is the coefficient of interest, $\mathbf{W}_1$ is a set of observed control variables, and $\mathbf{W}_2$ is a set of unobserved control variables. Equation (1) refers to the true model and returns an unbiased estimate of β . Equation (2) is the estimation equation for the results in Table 7 of the main paper, with $\mathbf{W}_1$ consisting of the full set of observed control variables. Estimates of $\tilde{\beta}$ will be biased unless $\mathbf{W}_2$ is uncorrelated with either X , Y , or both. Equation (3) is a naive model. Estimates of $\hat{\beta}$ will be more biased than those of $\tilde{\beta}$.

The Oster (2017) approach uses coefficient movements between the naive estimate ($\hat{\beta}$) and the controlled estimate ($\tilde{\beta}$) combined with movements in R-squared values to gauge the degree of potential omitted variables bias. Heuristically, estimates that move little with the inclusion of control variables that cause substantial increases in R-squared are indicative of limited omitted variables bias. More formally, the approach relies on two assumptions. The first assumption regards the so-called “coefficient of proportionality”, δ , which denotes the degree to which the observed controls ($\mathbf{W}_1$) determine treatment relative to the unobserved ($\mathbf{W}_2$). Setting $\delta = 1$ allows the unobserved controls to be as influential of the treatment as the observed controls. This assumption is likely to hold when the observed controls are among the strongest determinants of treatment.

The second assumption regards the maximum R-squared value (R_{max}^2) from the hypothetical estimation of Equation (1), the true model. The difference between R_{max}^2 and $R_{controlled}^2$ (from Equation (2)) determines the explanatory power of unobserved omitted variables after accounting for the observed control variables. In the presence of measurement error or idiosyncratic variation in the outcome, $R_{max}^2 < 1$.

Oster (2017) shows that with assumptions about R_{max}^2 and δ it is possible to use coefficient movements in β between the naive and controlled regressions to calculate the potential bias from omitted variables. This results in an identified set, bounded on one side by the controlled estimate and on the other by the bias-adjusted estimate, that contains the unbiased estimate. A result is deemed robust if the identified set excludes zero.

Note that using coefficient stability between Equations (2) and (3) to argue for causality is equivalent to arguing treatment is *unconditionally* exogenous: $\hat{\beta}$ varies little from $\tilde{\beta}$ because $\mathbf{W}_1$ does not confound. And because $\mathbf{W}_1$ does not confound, $\mathbf{W}_2$ is also unlikely to confound (especially when we believe $\mathbf{W}_1$ constitutes the strongest determinants of treatment). The framework can easily be extended to the case where treatment is believed to be exogenous only after conditioning on a set of control variables, $\mathbf{M}$. In this case, the variables in equations (1)-(3) are first residualized with respect to $\mathbf{M}$ (equivalently, $\mathbf{M}$ is included in equations (1)-(3)).

How to select conservative values for R_{max}^2 and δ ? Oster (2017) re-analyzes experimental studies to identify conservative values of R_{max}^2 and δ under which a non-zero bias-adjusted effect would be consistent with exogenous treatment assignment. These parameter values are then recommended as a robust reporting standard. The intuition of this test follows from the discussion above: observational studies implicitly argue that the treatment is exogenous. Including controls should not change the coefficient because there is no confounding. In experimental studies, this assumption is known to hold. Control variables will still

influence the coefficient estimate due to idiosyncratic imbalance across groups. Thus it is possible to use the stability of treatment estimates in randomized data as a guide to how much stability would be expected in observational data if the treatment were assigned exogenously. To do so, Oster (2017) draws on a large sample of randomized studies published in *American Economic Review*, *Quarterly Journal of Economics*, *Journal of Political Economy*, *Econometrica* and the *American Economic Journal Applied Economics* from 2008 through 2013.

Oster (2017) assumes the effects estimated in randomized data are causal and that they should therefore survive the bias-adjustment procedure. Robustness cutoff values are based on the value of R_{max}^2 and δ under which the bias-adjusted effect is distinct from zero in 90 per cent of experimental studies. This leads to the values of $R_{max}^2 = 1.3 \times R_{controlled}^2$ and $\delta = 1$. Substantively, this assumes unobservables explain as much of the variation in treatment as the observables and explain 30 per cent of the variation in the outcome explained by the included controls. For full details, see Oster (2017).

In the context of Table 7 in the main paper, the effect of interest is either the *Hardships Index* or Government or NGO outreach, $\mathbf{W}_1$ includes the full set of covariates reported in Table 7, including community fixed effects, and $\mathbf{W}_2$ is the set of all unobserved confounders. Our test is conservative in that we exceed Oster’s recommended standards for robustness by setting $R_{max}^2 = 2 \times R_{controlled}^2$ and $\delta = 1$ (rather than $R_{max}^2 = 1.3 \times R_{controlled}^2$ and $\delta = 1$). Substantively, this sets unobservables to be as influential as the full set of control variables (including fixed effects) in explaining both the outcome and treatment.

The results of this sensitivity analysis are presented in Appendix Table A.3. The first column shows the “naive” effect of either the hardships index of government or NGO outreach on the outcome, estimated from a bivariate regression. The second column presents estimates of the fully controlled effects, reported in Table 7 of the main article. The third and fourth columns show the bias-adjusted effect and identified set under Oster (2017)’s recommended standards for robustness ($R_{max}^2 = 1.3 \times R_{controlled}^2$ and $\delta = 1$). Under this level of confounding, the identified sets exclude zero in all three analyses. The fifth and sixth columns show the bias adjusted effect and identified set assuming $R_{max}^2 = 2 \times R_{controlled}^2$ and $\delta = 1$. Even under this level of confounding, the identified sets exclude across all three analyses. Substantively, the results of this exercise indicate that omitted unobservables would have to be substantially more confounding than observables to reduce the effects reported in Table 7 of the main paper to zero.

Table A.3: Oster Sensitivity Analysis

	Naive effect (Std. Error) [R2]	Controlled effect (Std. Error) [R2]	R2max = 1.3 x controlled R2		R2max = 2 x controlled R2	
			Bias-adjusted effect:	Identified Set	Bias-adjusted effect:	Identified Set
Effect of hardships index	-0.13 (0.02) [0.08]	-0.12 (0.02) [0.09]	-0.11	[-0.13, -0.11]	-0.09	[-0.13, -0.09]
Effect of Govt outreach	0.41 (0.05) [0.12]	0.39 (0.05) [0.14]	0.34	[0.34, 0.41]	0.24	[0.24, 0.41]
Effect of NGO outreach	0.14 (0.05) [0.09]	0.15 (0.05) [0.11]	0.15	[0.14, 0.15]	0.15	[0.14, 0.15]

Notes: The first column shows the “naive” effect of either the hardships index, government outreach, or NGO outreach on the trust in government index, estimated via bivariate regression. The second column presents estimates of the fully controlled effects, reported in Table 2 in the main article. The third and fourth columns show the bias-adjusted effect and identified set under Oster’s recommended standards for robustness ($R_{max}^2 = 1.3 \times R_{controlled}^2$ and $\delta = 1$). The fifth and sixth columns show the bias adjusted effect and identified set assuming $R_{max}^2 = 2 \times R_{controlled}^2$ and $\delta = 1$.